

Answer_Assignment 3_Level 3_Seating Arrangement

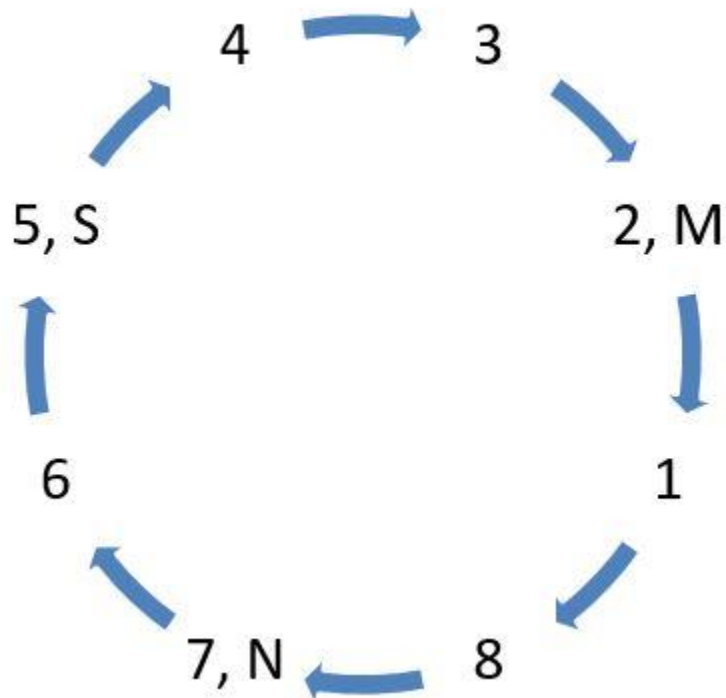
I.

1. 2
2. 5
3. 5
4. 3
5. 2

Solution:

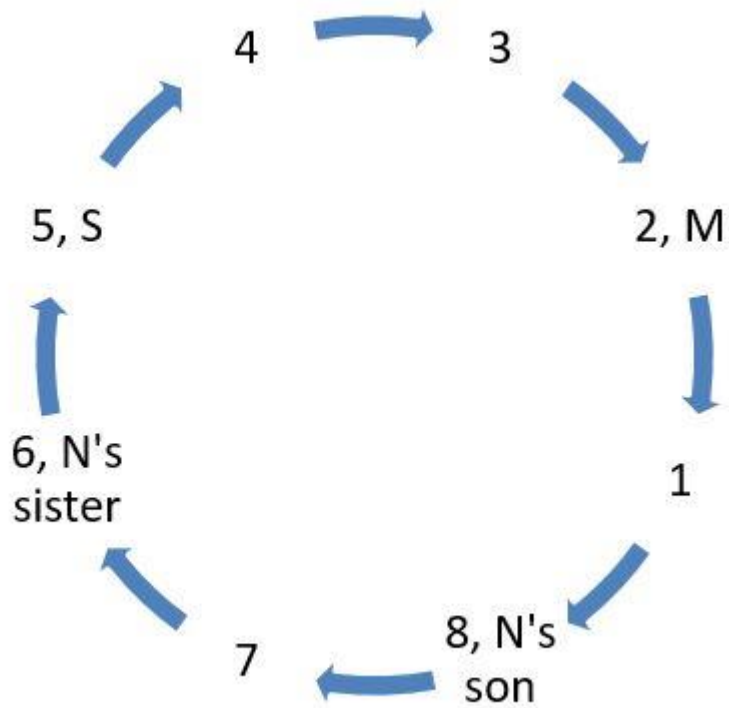
2 people sit between M and S. Lets fix the position of S first. And then we can place M accordingly. Fixing S at 5, we get 2 possibilities for M.

Case (i) M at 2

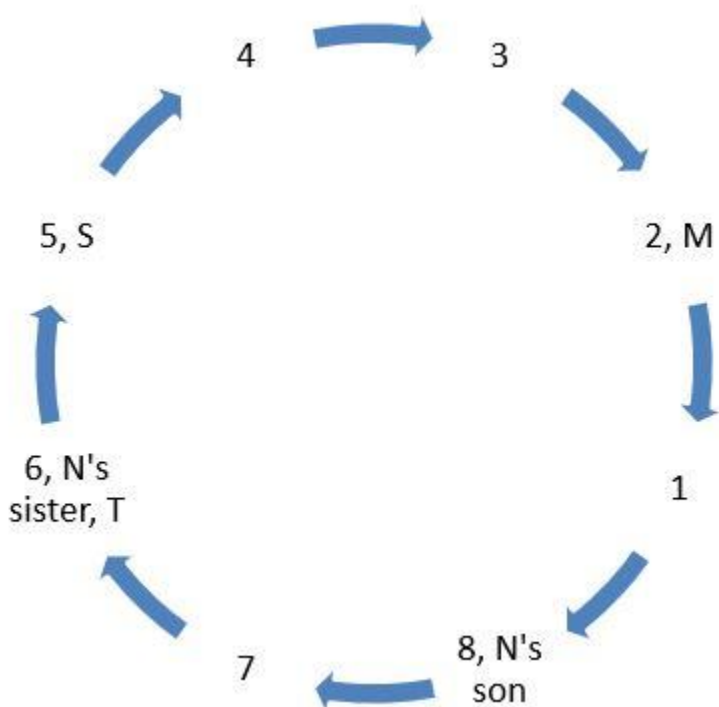


From i, ii

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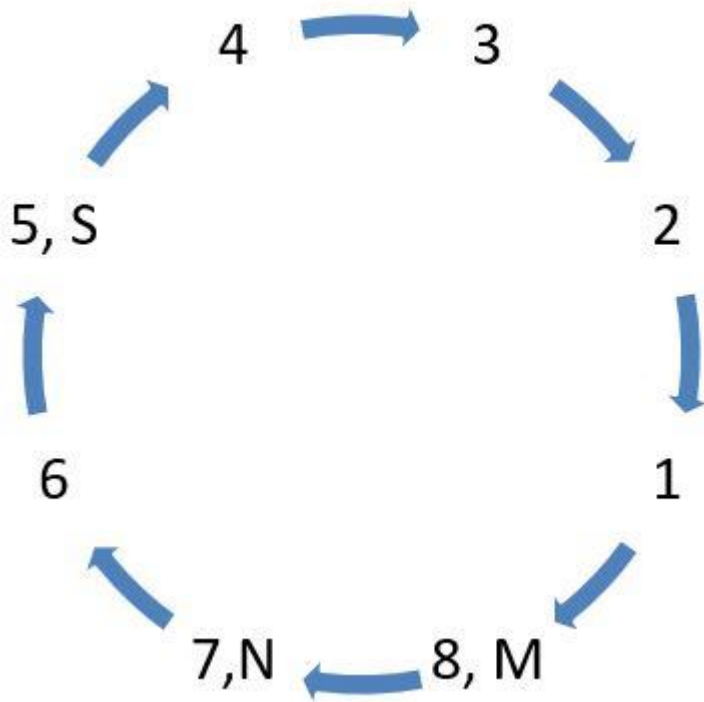
From iii,



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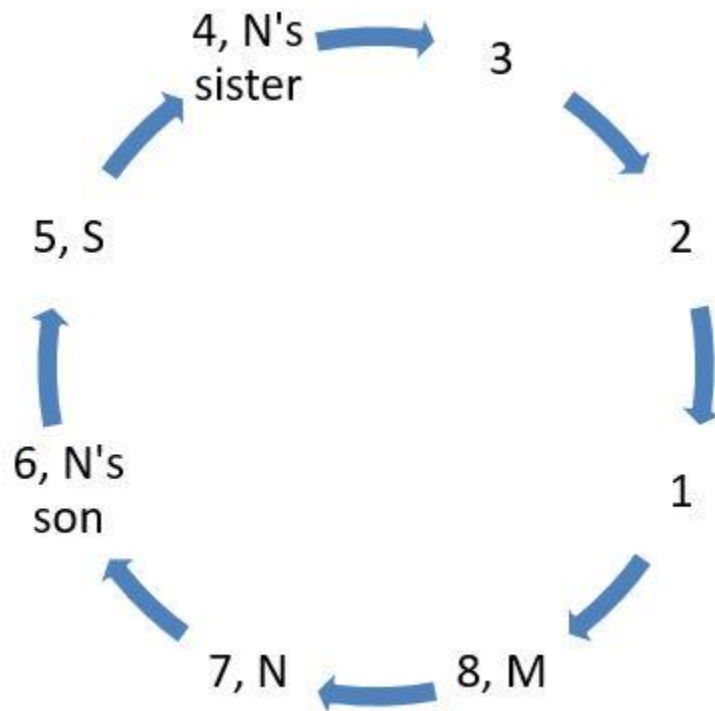
But K sits to the immediate right of S according to iii, which is not possible here. thus, case (i) discarded.

Case (ii), M at 8,



From i,ii,

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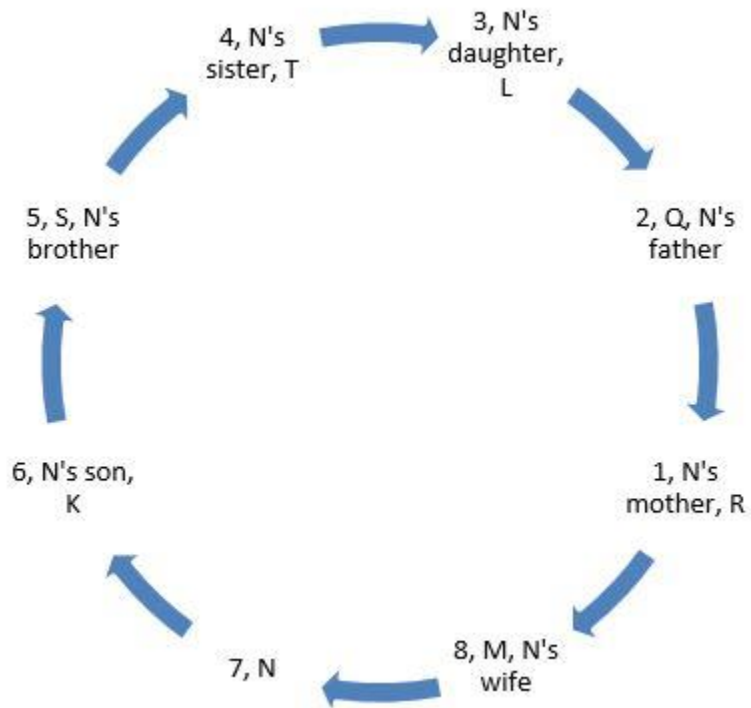


From iii and iv, we get the following arrangement. But also T is an immediate neighbour of n's brother. N's brother can either be at 3 or 5.

If N's brother is at 3, according to iv and v, N's mother should be seated at 7 but it is already occupied by N. Thus this case is discarded.

If N's brother is at 5, then, according to iv and v, at 1 will be N's mother.

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Family tree

R(-) X Q(+)

|

N(+) X M(-) = S(+) = T(-)

|

L(-) = K(+)

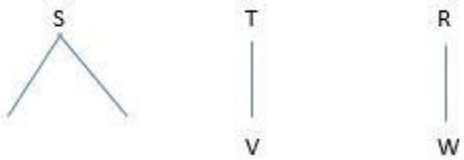
II.

1. 2
2. 2
3. 4
4. 4
5. 1

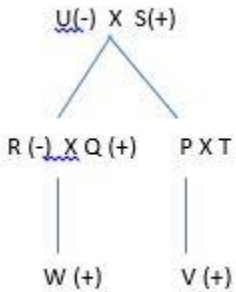
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SOLUTION:

From (i),



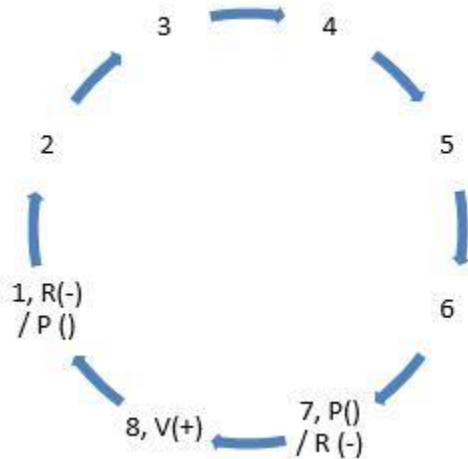
From (ii) and (iii) and since no unmarried females, W and V are males.



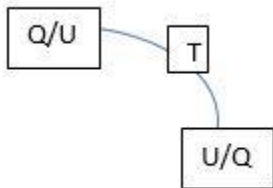
Fix V's position at 8.

This means, according to (iii), R and P can occupy positions 7 and 1.

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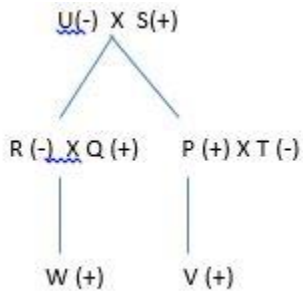
And Also according to (iii), T sits exactly between Q and U.



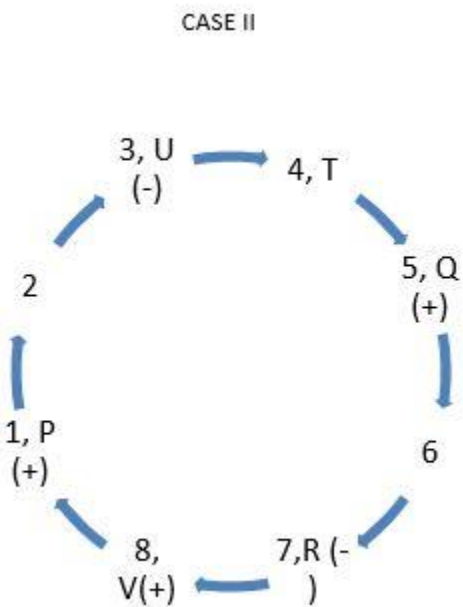
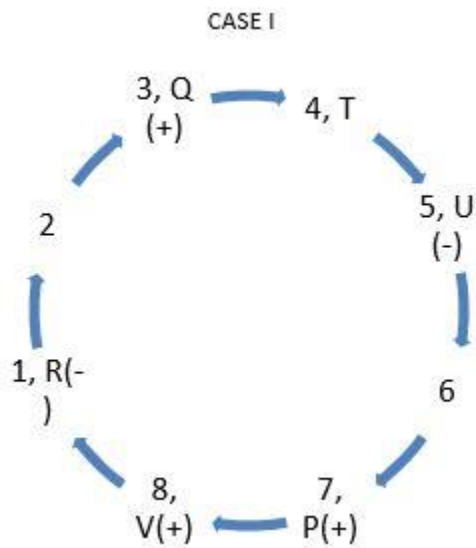
Also, Only one person sits between R and Q but he is not W.

So, two cases arise. Since in both the cases, Q will sit opposite P, P has to be a male as according to (iii), Q does not sit opposite a female.

Family tree is



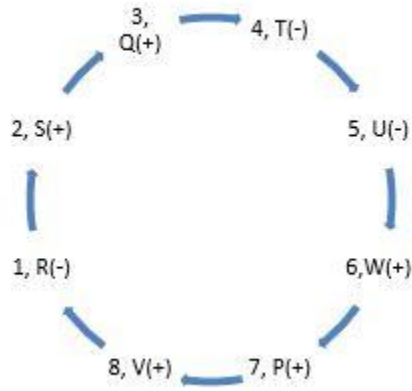
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As according to (iv), on the immediate left of S has to be a son of S (P or Q), case II will be discarded.

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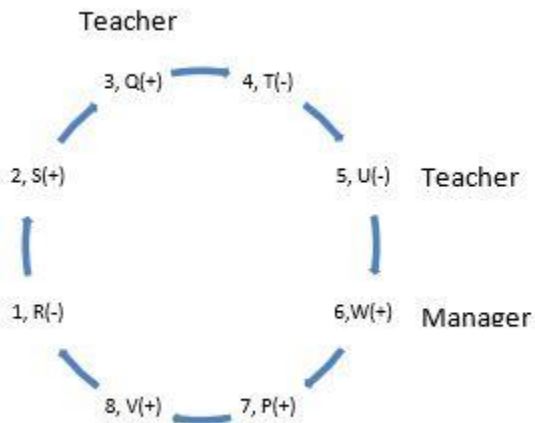
According to (iii), W does not sit between Q and R, thus, W will be at 6 and S will be at 2.



Now, there have to be

- 3 – Artists
- 2 – Athletes
- 2 – Teachers
- 1 – Manager

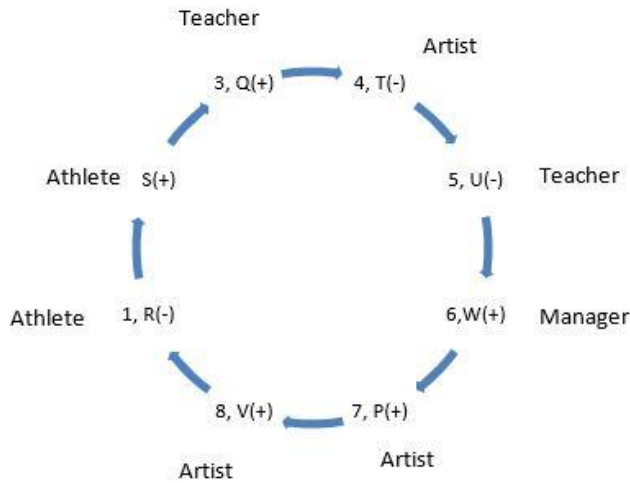
Since from (vi), Q and U are not Athletes or Artists, they are definitely Teachers and also we come to know that W is a Manager.



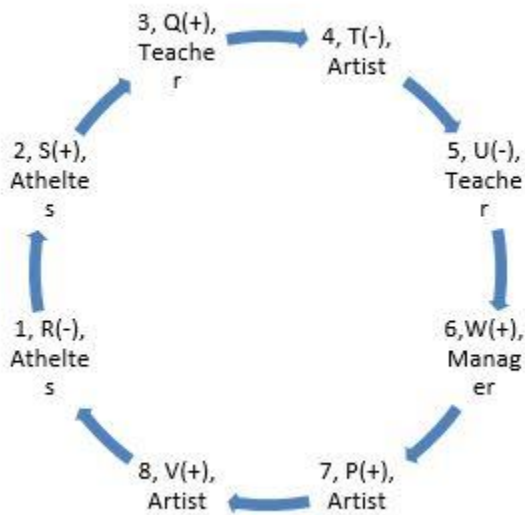
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We got 2 teachers, 1 manager and also from (v), it is clear that R and S cannot be artists, thus they are athletes.

Now, remaining three, T, V and P are Artists.



Final Solution:



III.

1. 1
2. 3

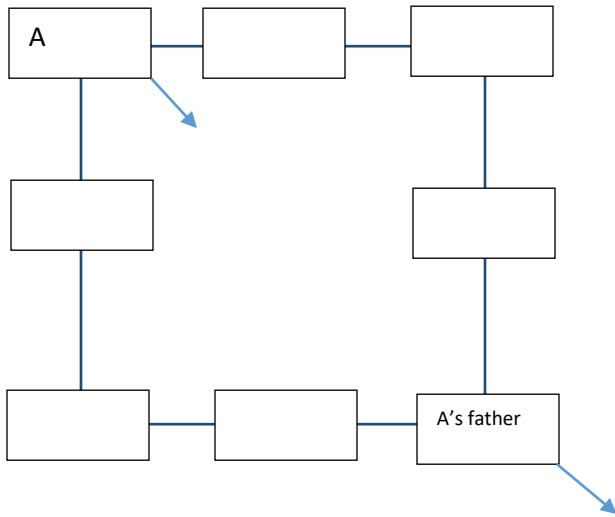
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3. 4
4. 1
5. 4

Solution:

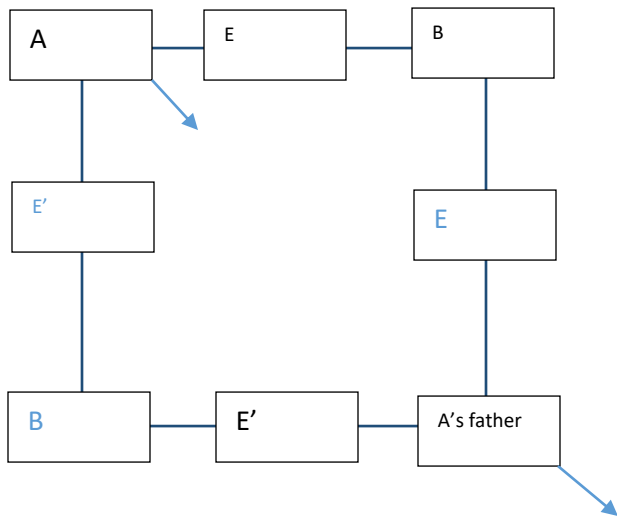
From 4,

Case I – A sits at a corner



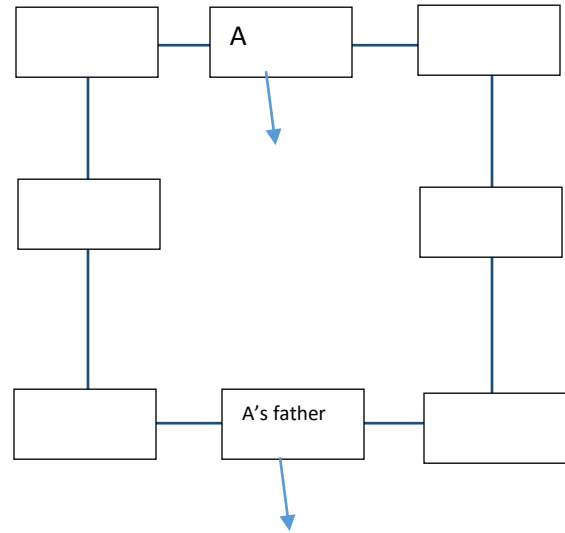
From 7,

Case I – A sits at a corner

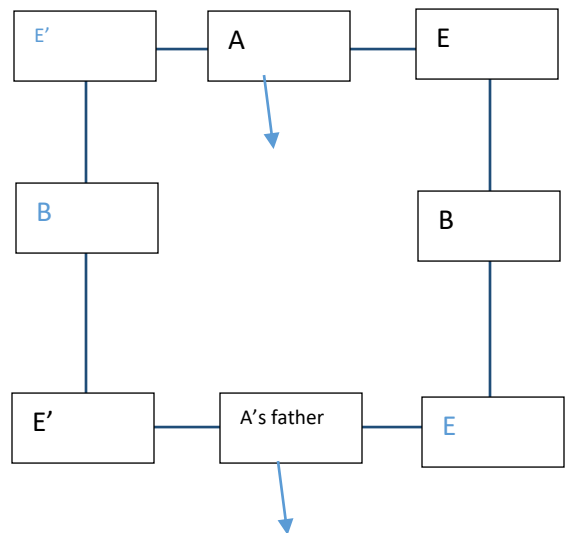


From 8, we know, B sits at one of the centers, thus case I discarded.

Case II – A sits at a center

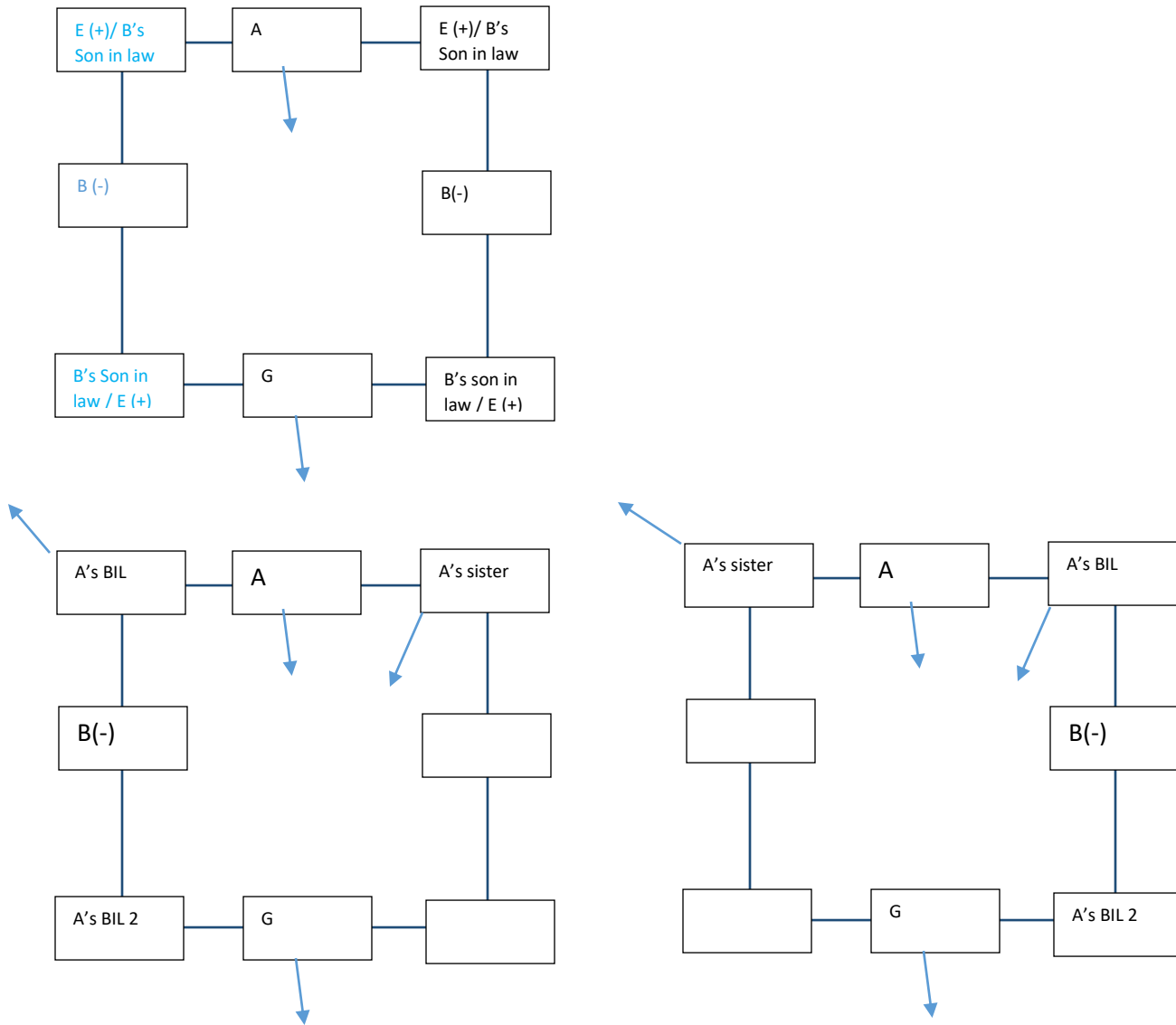


Case II – A sits at a center



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From 2, A sits exactly between his sister and brother in law. This means A is male having a married sister. But since there are only three generations in a family, B has two son in laws, this means A has two sisters



Since A sits next to his sisters, and B has two son in laws, this means B has at least three children, one son and two daughters. Both the daughters married.

From 7 we know that E is the son in law of G, thus G is a male because mother in law we know is B. Also, E is D's father.

B(-) X G (+)

|

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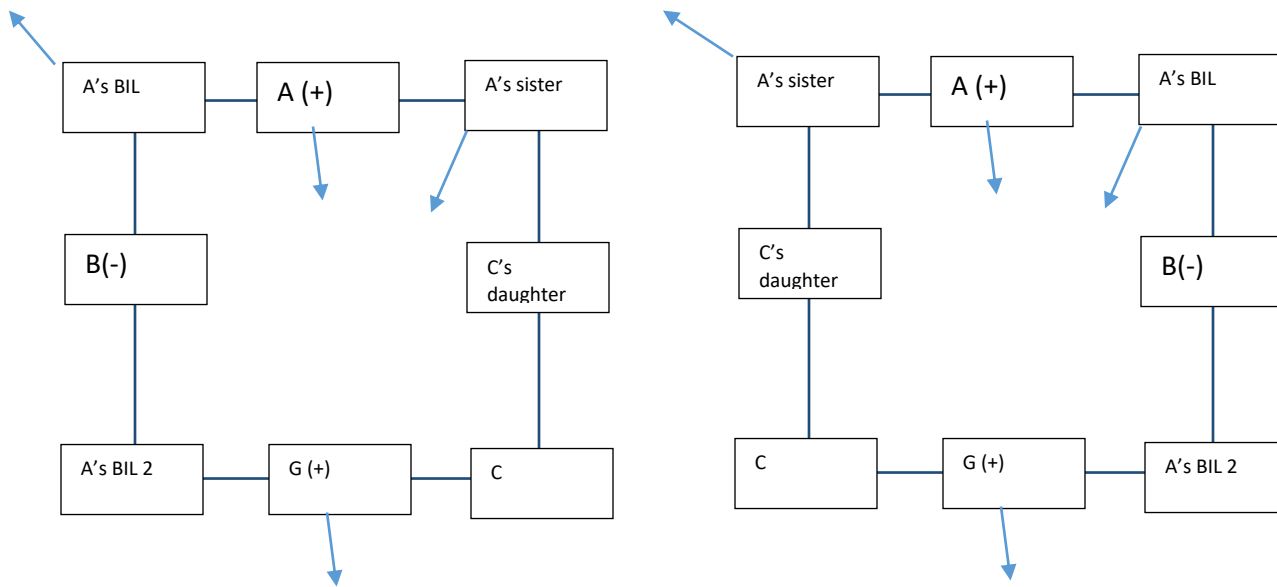
$$(+) \times (-) = A (+) = (-) \times E (+)$$

I

D

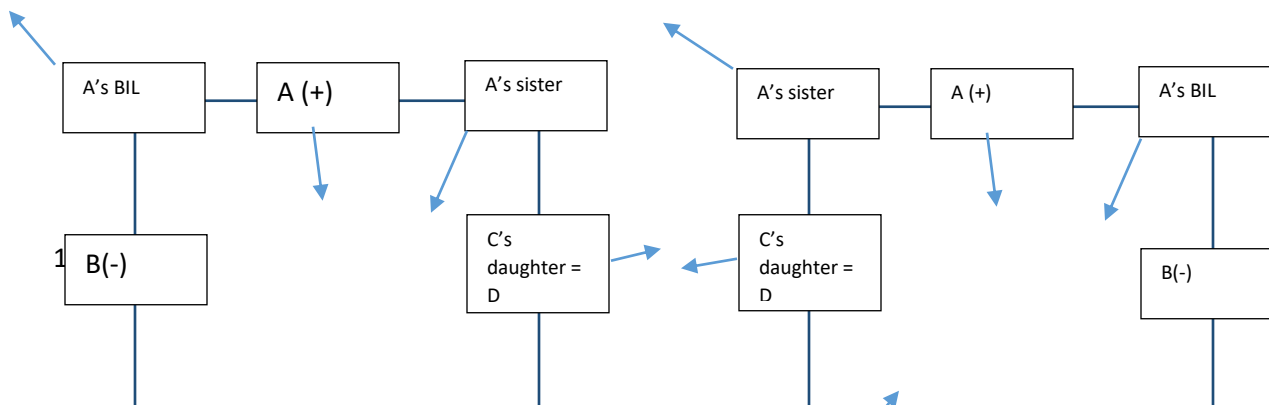
In the family tree, 8 positions are filled.

From 2,



From 6, D sits to the immediate left of her mother who is facing towards the center and who in turn sits third to the right of her mother. H faces in direction opposite to that of his wife and his mother in law.

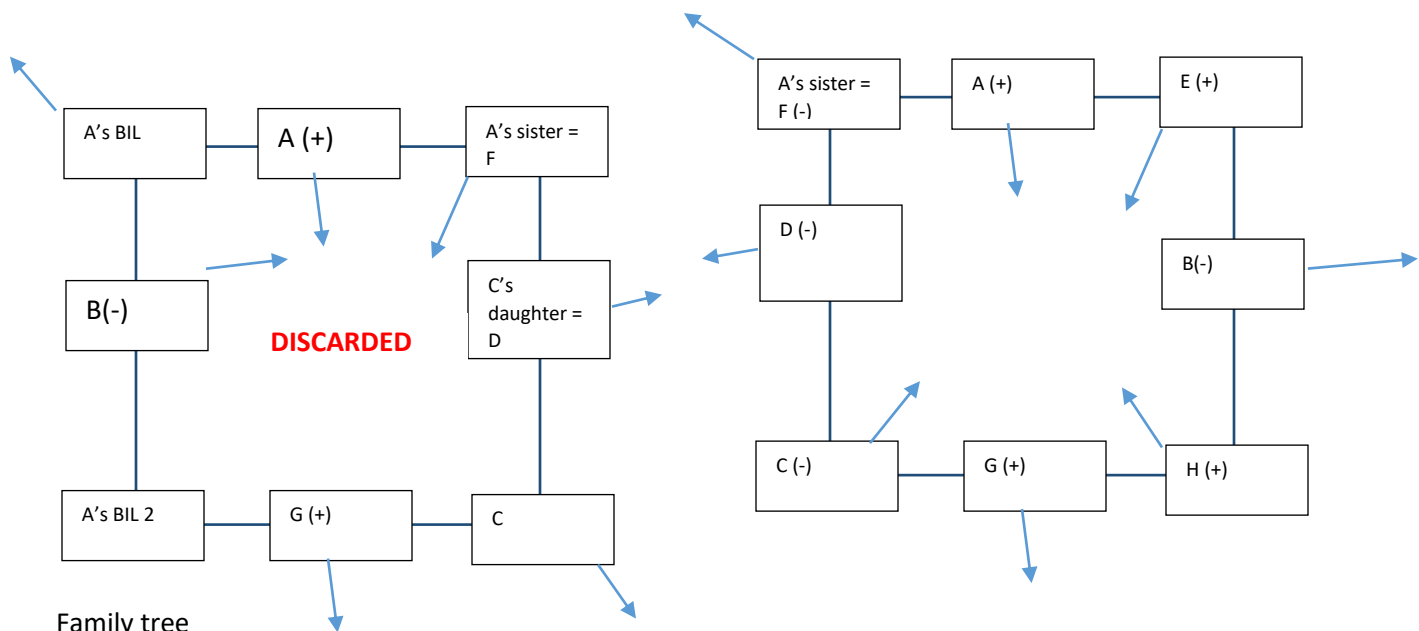
From 2,



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Both the neighbours of C face the same direction.

F sits to the immediate right of her niece, but in case I, to the immediate right of D (F's niece) is her mother. Thus case I discarded. Also, E sits exactly between A and B. Also, H faces direction opposite to his wife and Mother in law. Thus,



Family tree

B(-) X G (+)

|

H (+) X F (-) = A (+) = C (-) X E (+)

|

D (-)

IV.

Solution:

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Case I (Shilpa facing inside)

Case II (Shilpa facing outside)

From 6, we know Shilpa is from either Delhi or Mumbai. Also, Chitra & Mehak not from Manipal.

Persons	City								Company						
	Chen	Mani	Del	Mum	Hyd	Sikk	Chand		Capgemini	Wip	Acc	iGATE	Info	Amaz	Flip
Mehak		X								X					
Prerna										X					
Shilpa	X	X			X	X	X			X					
Ekta										X					
Shruti										X					
Garima		X					X		X	Y	X	X	X	X	X
Chitra		X								X					

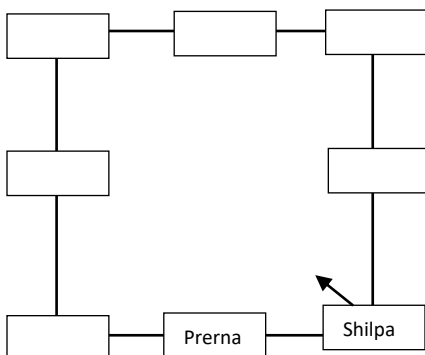
The one who is working in Capgemini sits immediate right of the one who is working in Wipro i.e Garima.

Capgemini Garima (Outside) Sikkim Uni

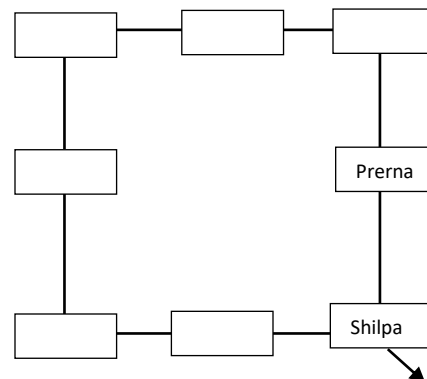
OR

Sikkim Uni Garima (Inside) Capgemini

From 7, The one from Mumbai City sits 3rd to the right of Prerna.



Case I (Shilpa facing inside)

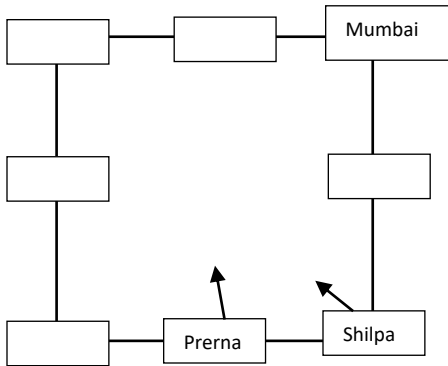


Case II (Shilpa facing outside)

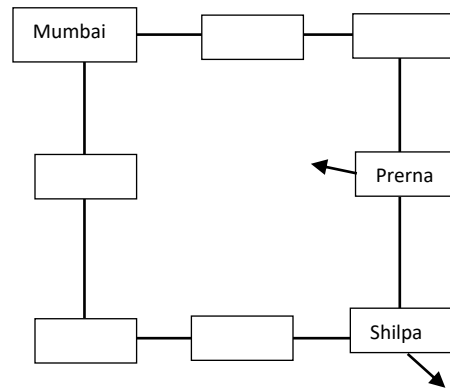
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Prerna may face either inside or outside the table. Thus, forming two more sub cases of each case I and case II

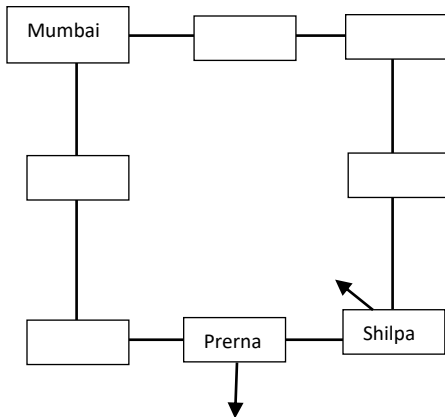
Case I (a) – Prerna facing inside



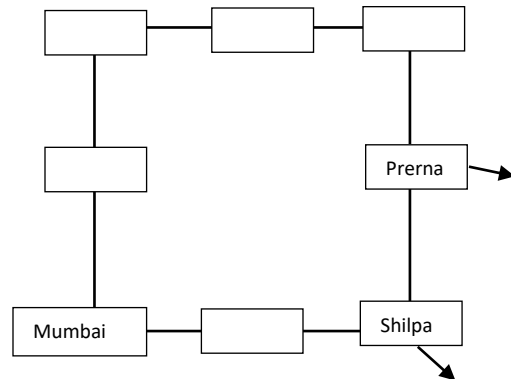
Case II (a) – Prerna facing inside



Case I (b) – Prerna facing outside

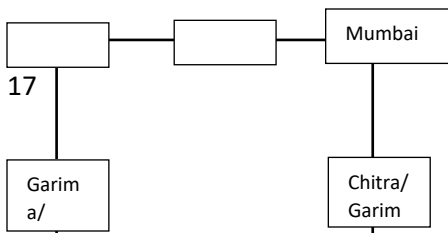


Case II (b) – Prerna facing outside

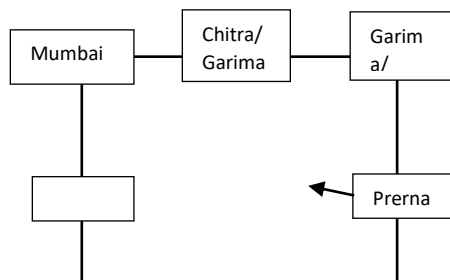


Also, Garima and Chitra sit opposite each other. Also, from 9th we know, the person from Mumbai sits diagonally opposite to the one who is working in Infosys.

Case I (a) – Prerna facing inside

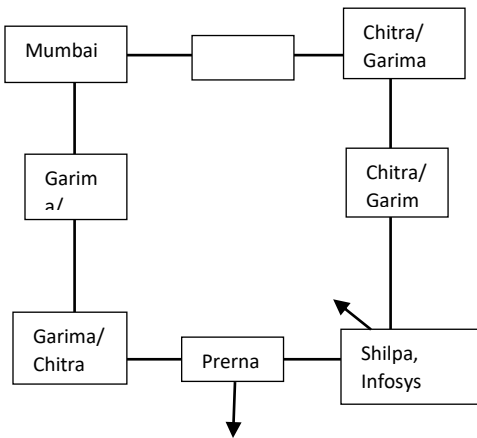


Case II (a) – Prerna facing inside

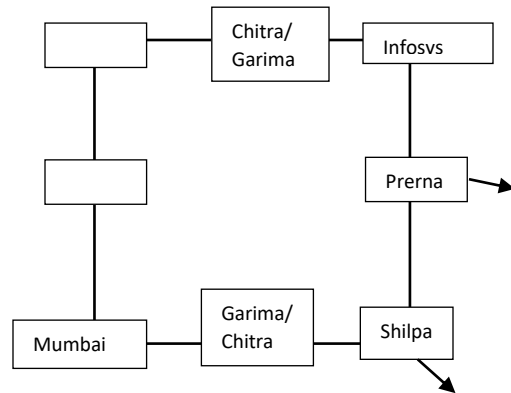


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Case I (b) – Prerna facing outside



Case II (b) – Prerna facing outside



In no case, Shilpa is from Mumbai City. Thus **Shilpa is from Delhi City.**

From 5 and 7 we know,

Ekta faces the direction similar to that of Chitra but opposite to that of Mehak.

But, Garima and Chitra face away from each other, which means

Chitra faces outside, thus Ekta faces outside and Mehak faces inside. Also, Garima faces outside.

And from 8, the one from Manipal city is working in Amazon, thus the ones not from manipal city cannot do Amazon.

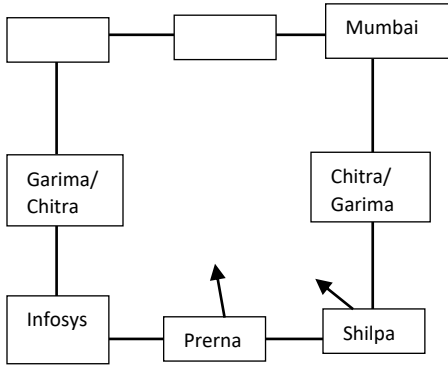
Persons	City							Company						
	Chen	Mani	Del	Mum	Hyd	Sikk	Chand	Capgemini	Wip	Acc	iGATE	Info	Amaz	Flip
Mehak		X							X				X	
Prerna									X					
Shilpa	X	X			X	X	X		X				X	
Ekta									X					
Shruti									X					
Garima		X					X	X	Y	X	X	X	X	X
Chitra		X							X				X	

We know that Ekta, Shruti and Mehak sit on one side with Shruti in the middle.

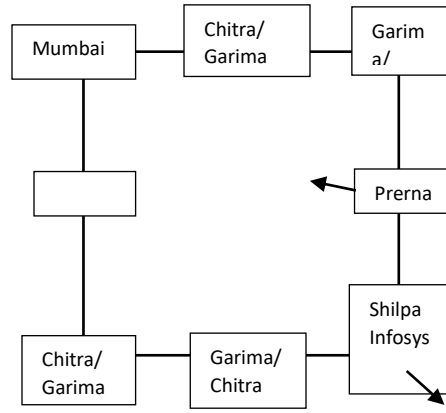
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Thus, the only possibilities for thAccenture to sit on one side are:

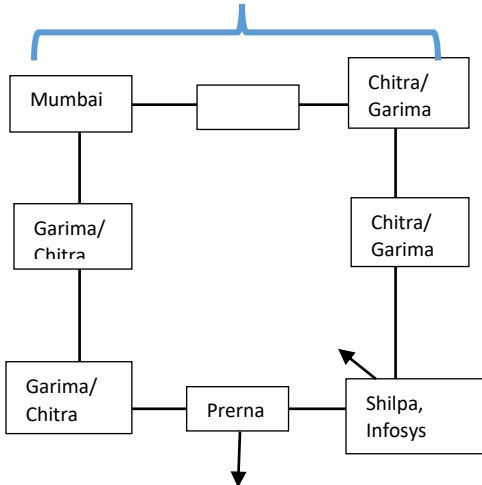
Case I (a) – Prerna facing inside



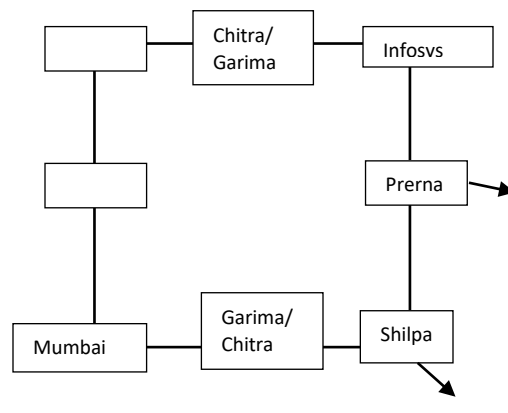
Case II (a) – Prerna facing inside



Case I (b) – Prerna facing outside

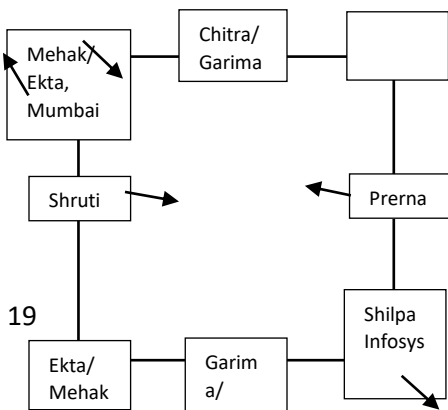


Case II (b) – Prerna facing outside

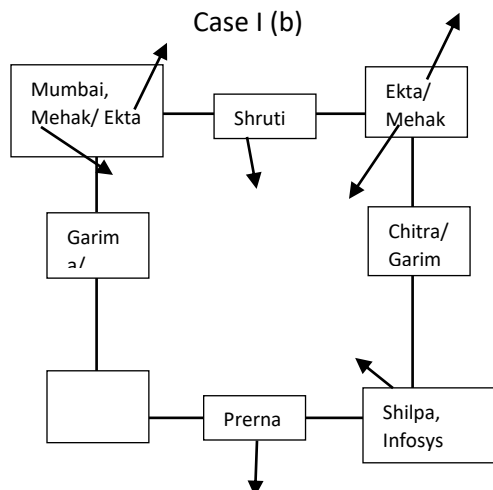


If in case I (a) and Case II (b) Ekta, Shruti and Mehak occupy the side highlighted, Then no seat rAccentureains vacant (all 8 occupied), thus case I (a) will be discarded. Thus proceeding with cases I (b) and II (a)

Case II (a)



Case I (b)

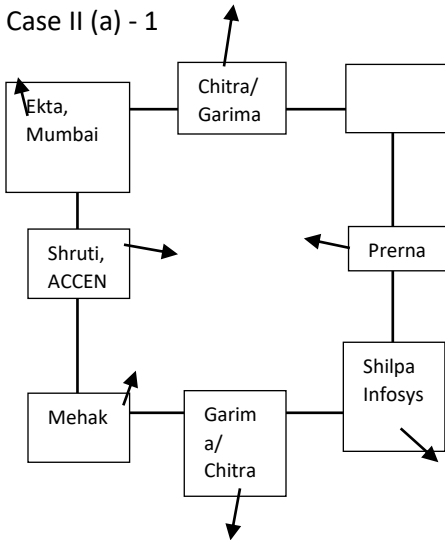


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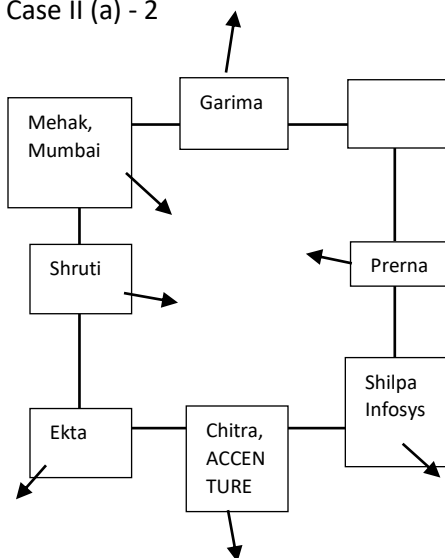
The one sitting on immediate left of Ekta is working in Accenture. In case II (a)

Case II (a) - 1

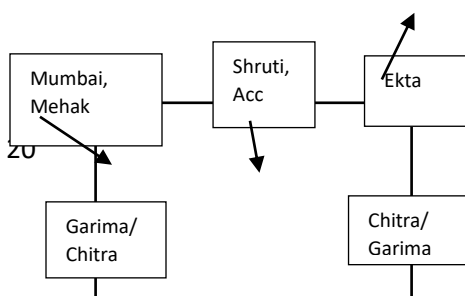


Case II (a) – If Mehak goes to Mumbai, Ekta sits next to Chitra/Garima. But, Garima is not working in Accenture, Thus, Chitra is seated next to Ekta and Garima is seated next to Mehak

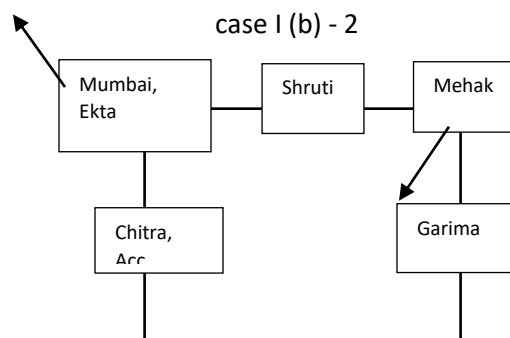
Case II (a) - 2



Similarly, in case I (b) – 1



case I (b) - 2

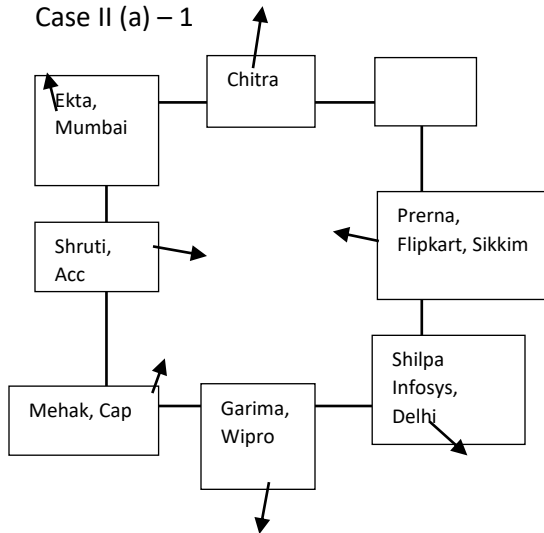


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The one who is working in Capgemini sits immediate right of the one whom working in Wipro i.e. Garima

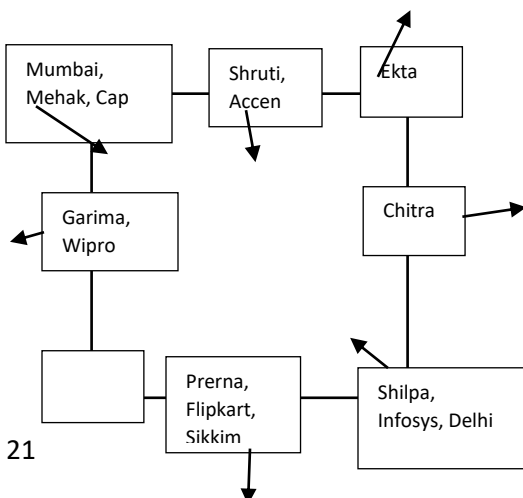
Case II (a) – 1



The only people possible for company in manipal city (refer table), are Perna and Ekta but city are already allotted to these two people in this case, thus this case will also be discarded.

Case II (a) - 2 Discarded as the place right of Garima needs to be occupied, but cannot be in this case as it is the only vacant seat. Similarly, Case I (b) – 2 also discarded.

Case I (b) - 1



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Perna is allocated Sikkim City, Thus Ekta works in Amazon in Manipal. Filling in missing entries.

